Do Styles Make Fights?

Data-Driven Style Discovery, Outcome Prediction, and Upset Detection in the UFC

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Abstract

"Styles make fights" is the most-repeated cliché in mixed martial arts. Statistical models of UFC outcomes almost never encode style explicitly: they regress wins on per-minute strike and grapple rates together with physical and experiential covariates, and whatever residual structure style contributes is left to the classifier to exploit. This thesis asks whether the cliché survives a rigorous test—whether latent fighting styles can be recovered directly from per-fight data, and whether, once recovered, they sharpen our predictions of *who* wins, *how* the fight unfolds, and *when* a favourite is vulnerable. Using a leakage-free pipeline over roughly seven thousand UFC bouts through 2024, we learn soft Gaussian mixture posteriors at $K = 3$ and $K = 5$ and a three-dimensional autoencoder embedding and validate them against a curated panel of twenty-five champions: Khabib Nurmagomedov, Islam Makhachev, and Georges St-Pierre all embed in the takedown-heavy wrestler cluster, while Max Holloway, Jose Aldo, Conor McGregor, Israel Adesanya, and Alex Pereira all embed in the active distance-striker cluster.

The answers to the three questions differ sharply in kind. For *who wins*, a stack of six model families on the full non-market feature set lifts held-out AUC from 0.600 to 0.670 above a Z-score baseline, and a version that includes Vegas as a feature edges the market from AUC 0.751 to 0.775 with a paired bootstrap 95% CI on the difference that excludes zero on all three of AUC, Brier, and log loss ($p = 0.04,\ 0.02,\ 0.01$); a paired null finding is that hybrid generalists do not systematically outperform specialists on win rate. For *how the fight unfolds*, top-quintile autoencoder style divergence raises the observed finish rate from ≈ 0.47 to ≈ 0.56, and a hierarchical $P(\text{finish}) \cdot P(\text{method} \mid \text{finish}) \cdot P(\text{winner} \mid \text{method})$ decomposition cuts six-way log loss from 2.15 to 1.94. For *when a favourite is vulnerable*, the autoencoder group is the largest non-market contributor to a residual detector $r = p_{\text{model}} - p_{\text{vegas},A}$, and a de-vigged flat-stake ROI simulation returns +17.5% on 411 bets at $\theta = 0.15$ with a bootstrap 95% CI of [+1.5%, +34.1%]. Style, in short, does help "make fights," but in a more specific sense than the cliché is usually stated: it helps least where MMA analysts most often invoke it (who wins) and most where they rarely do (how fights end and when markets are wrong).

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Chapter 1

Introduction

"Styles make fights" is the most-quoted cliché in mixed martial arts. Broadcasters explain outcomes through the language of archetypes—striker against grappler, pressure fighter against counter-puncher—and attribute results to the stylistic pairing rather than to raw quality. Statistical work on UFC outcomes has been slow to encode this intuition. The standard pipeline regresses wins on per-minute striking rates, takedown accuracy, control time, physical attributes, and recent form (Hitkul et al. 2019; Yan et al. 2024; Yin 2024); each feature enters as an independent input and whatever structure "style" contributes is left to the downstream classifier to discover implicitly. When style does enter explicitly, it does so through hand-applied labels—"striker," "grappler," "hybrid"—that analysts disagree about and that cannot represent continuous variation in fighting behaviour. This thesis asks whether latent fighting styles can instead be recovered directly from per-fight data, and whether, once recovered, they sharpen our predictions of *who* wins a fight, *how* the fight unfolds, and *when* a favourite is vulnerable.

The three questions are ordered by the difficulty with which they can be answered in the affirmative. Predicting wins in a combat sport with a roughly 55% finish rate and large fighter-to-fighter variance is hard, and the betting market already integrates much public and private information. Predicting how a fight unfolds is a richer task with more outcome categories in which style can leave a fingerprint. Identifying upsets is easier still if genuine style effects are concentrated in the tails of mismatched matchups—where a favourite looks dominant by every numeric feature but is stylistically exposed. The rest of the thesis finds that this ordering is exactly right. For *who wins*, a stack of six model families on the full non-market feature set lifts held-out AUC from 0.600 to 0.670 above a Z-score baseline and, with de-vigged Vegas probabilities added as a feature, edges the market itself by a small but statistically significant margin. For *how fights unfold*, top-quintile autoencoder

style divergence raises the observed finish rate by roughly eight percentage points, and a hierarchical method-of-victory decomposition cuts six-way log loss by $\sim 10\%$ versus a flat classifier. For *when a favourite is vulnerable*, a Vegas-aware residual upset model in which the autoencoder embedding is the single largest non-market contributor returns a de-vigged ROI of +17.5% on 411 simulated bets at threshold $\theta = 0.15$ with a paired bootstrap (Efron 1979) 95% CI of [+1.5%, +34.1%]. Style, in short, does help make fights, but in a more specific sense than the cliché is usually stated: it helps least where MMA analysts most often invoke it (who wins) and most where they rarely do (how fights end and when markets are wrong).

Chapter 2

Background and Related Work

The peer-reviewed literature on supervised UFC outcome prediction is small. Hitkul et al. (2019) present an early comparative study that evaluates random forests (Breiman 2001), decision trees, a perceptron, stochastic-gradient-descent logistic regression, support vector machines, and k-nearest neighbours on a UFC scrape, reporting roughly 63% accuracy on a random hold-out. More recently, Yan et al. (2024) benchmark random forests, gradient boosted decision trees, and SVMs on the Kaggle "UFC-Master" dataset and identify age difference and fight date as their most influential features. Closest in spirit to the present work, Yin (2024) apply factor analysis to per-fight statistics, cluster fighters into a handful of technical styles with k-means, and feed the style assignment into an ensemble of logistic regression, random forest, SVM, XGBoost (Chen and Guestrin 2016), and a shallow neural network, reaching 65.5% accuracy and showing in an ablation that the style factor contributes positively. None of these studies use a chronological split, a calibrated probabilistic baseline, or a continuous style embedding, and none compare their predictions to market odds. A recurring pitfall across the body of work is *data leakage*: career-aggregate features computed on the full dataset yield optimistically biased test metrics when the aggregation window includes the fight being predicted. The present thesis avoids this by computing rolling pre-fight statistics walked in strict chronological order (Chapter 3).

Two rating systems appear as features throughout. Elo (Elo 1978) updates each competitor after every bout based on the actual versus expected outcome; the rating difference, passed through a logistic CDF, yields a win probability. Glicko-2 (Glickman 1999, 2012) extends Elo with a rating deviation and a volatility term and behaves better for fighters with layoffs or small samples. Both are walked chronologically, so the rating used to predict a fight is strictly the rating before it resolves.

For style discovery we use two complementary representations. Gaussian mixture models

(Bishop 2006; Dempster et al. 1977; Hastie et al. 2009) are the soft-clustering counterpart of K-means: each fighter is associated with a posterior $\pi = (\pi_1, \ldots, \pi_K)$ over components rather than a hard label, so hybrid fighters are represented by a spread posterior rather than by an arbitrary hard assignment; the Shannon entropy $H(\pi) = -\sum_k \pi_k \log \pi_k$ summarises a fighter's "genericity" in a single scalar (the *Hybrid Score*). A feed-forward autoencoder (Hinton and Salakhutdinov 2006) implemented in PyTorch (Paszke et al. 2019) and optimised with Adam (Kingma and Ba 2014) embeds each qualified fighter into $\mathbb{R}^3$; differences and means of the two corners' embeddings enter downstream models as matchup features. The direct transfer of GMMs and autoencoders to per-fighter MMA profiles is, to our knowledge, novel.

Downstream, probability calibration matters whenever a model is used for betting or resource allocation. We use isotonic regression (Niculescu-Mizil and Caruana 2005; Zadrozny and Elkan 2002) both as a post-hoc calibrator on individual models and as the meta-learner in a stacked ensemble (Breiman 1996; Wolpert 1992); calibration quality is reported as expected calibration error (Guo et al. 2017) alongside the Brier score (Brier 1950). For significance testing on paired metrics we use the nonparametric paired bootstrap (Efron 1979), which yields percentile CIs and two-sided p-values without parametric assumptions. For interpretability we report *grouped* permutation importance (Breiman 2001) over semantic feature families—rolling rates, style posteriors, embeddings, ratings, physicals, Vegas—shuffling an entire group rather than individual columns, which is more legible than SHAP when group members are colinear. All pipelines are implemented in scikit-learn (Pedregosa et al. 2011), with XGBoost (Chen and Guestrin 2016) and LightGBM (Ke et al. 2017) used where noted.

Chapter 3

Data and Preprocessing

The primary dataset is the *UFC Fights, Fighters, and Events* Kaggle scrape of UFCStats (Alibi 2024): fight-level metadata (date, method, round, duration), per-fight positional striking and grappling statistics, and fighter-level attributes (date of birth, height, reach, stance, weight class). After cleaning, the merged table contains roughly seven thousand fights from 1993 through 2024. For the Vegas baselines we use a community-maintained betting-odds dump (Kaggle contributors 2024), filtered to US moneylines and one row per fight, with de-vigged implied probabilities $p_{\text{vegas},A} = o_A^{-1}/(o_A^{-1} + o_B^{-1})$. Vegas coverage is partial ($\approx 40\%$ of all fights, $\approx 60\%$ of test-era fights), so every Vegas-using analysis below is restricted to the Vegas-aligned subset.

The preprocessing pipeline is built around a rolling pre-fight aggregator that snapshots each fighter's cumulative career averages *before* incorporating the current bout. Output columns include pre-fight per-minute rates (significant strikes, takedowns, submission attempts, control ratio, positional striking shares), KO and submission rates for and against, and five-fight recent form. Because the aggregates at fight t strictly exclude fight t and all subsequent fights, rolling columns are never contaminated by the outcome they are used to predict—the single most common optimistic-bias bug in published MMA models. Per-minute rates are Z-scored *within weight class* to remove division-specific baselines. Elo ratings are walked with a standard specification (initial 1500, $K = 24$, logistic scale 400; Elo 1978) and Glicko-2 with monthly rating periods and $\tau = 0.5$ (Glickman 2012); pre-fight (R_A, R_B), their difference, and Glicko-2 rating deviations are attached as features.

A single artefact, `ufc_matchup_features.csv`, gathers every column used downstream: identifiers, rolling rates for both corners plus δ and μ transforms, K=3 and K=5 GMM posteriors and Hybrid Scores, the three-dimensional autoencoder embedding, physicals and weight-class one-hots, Elo and Glicko-2 ratings, an empirical style-pair heatmap probability

$p_{\text{heatmap},A}$ (train-only win rate of A's K=5 cluster versus B's), and $p_{\text{vegas},A}$ where available. A `split` column $\in \{\text{train}, \text{val}, \text{test}\}$ from an event-ordered 60/20/20 cut ensures every downstream model shares the same training, validation, and test events. All test-slice metrics are reported on the most recent 20% of events; model selection and isotonic calibration happen on `val`; cross-validation within `train` uses walk-forward folds. Every training row is augmented with its A/B mirror so models cannot exploit the arbitrary corner convention, and test metrics are reported on unmirrored rows so each fight counts once.

Figure 3.1 anchors what follows. Across the matchup table, significant-strike-volume delta is the strongest raw correlate of the outcome ($\rho \approx 0.19$); reach and height deltas carry weaker signal ($\rho \approx 0.07$ and 0.03); and a naïve hand-engineered "style clash" score correlates at essentially zero with who wins. This foreshadows the thesis's central pattern: style will matter more for *how* a fight unfolds than for the top-line outcome.

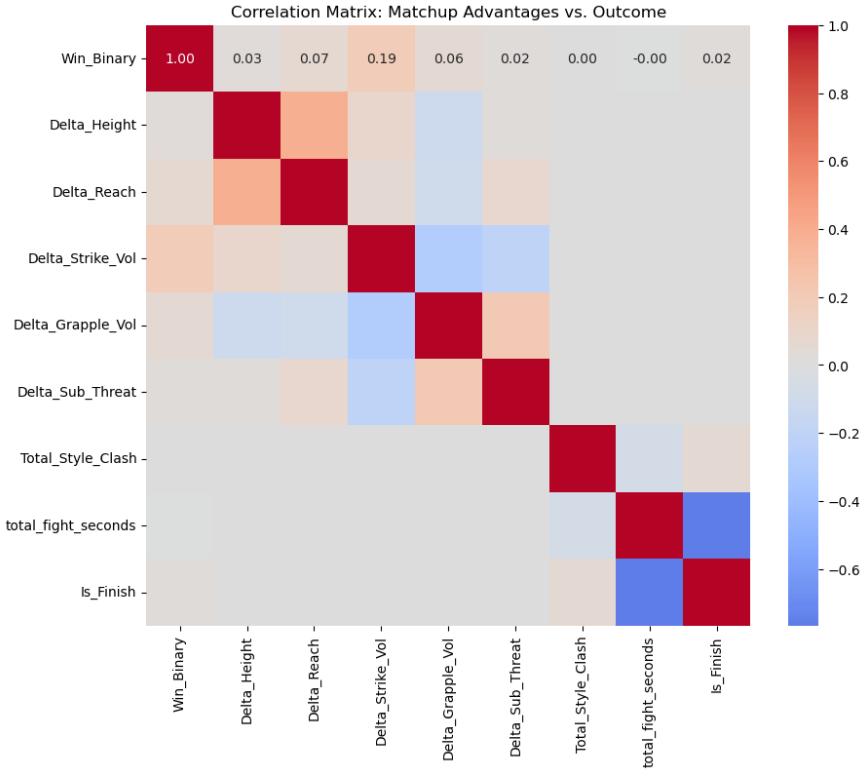

Figure 3.1: Matchup-level correlation matrix of A−B deltas with the win and finish indicators. Strike-volume delta is the largest raw correlate of winning ($\rho = 0.19$); the "style clash" score is near zero with the outcome but, as Chapter 6 shows, carries substantial signal for the method of victory. Source: notebook 07.

Chapter 4

Discovering Fighting Styles

Style discovery operates on a fighter-level profile $x \in \mathbb{R}^7$ summarising each qualified fighter's career: the Z-scored per-minute rates of significant strikes, takedown attempts, and submission attempts, the control-time ratio, and the three positional shares of striking (distance, clinch, ground). Fighters with fewer than five UFC bouts are excluded as too noisy to support a stable assignment. The same seven features feed all three representations below—K-means, Gaussian mixtures, and an autoencoder—so that any difference between representations reflects model choice rather than feature engineering.

4.1 GMM posteriors and autoencoder embeddings

A K-means baseline with $K \in \{4, 5, 6\}$ selects $K = 5$ by silhouette (Rousseeuw 1987) and serves downstream only as the backing label set of the style-pair heatmap below. The representations we actually use in Chapters 5–8 are soft. Gaussian mixtures with full covariance are fit at $K \in \{2, \ldots, 8\}$ and selected by AIC/BIC (Schwarz 1978); we persist $K = 3$ and $K = 5$ posteriors $\pi = (\pi_1, \ldots, \pi_K)$ via the EM algorithm (Dempster et al. 1977) together with the Hybrid Score $H(\pi) = -\sum_k \pi_k \log \pi_k$. Low H flags specialists with near-degenerate posteriors; high H flags generalists whose mass is spread across components. A feed-forward autoencoder $7 \to 32 \to 3 \to 32 \to 7$ with ReLU activations, trained with Adam (Kingma and Ba 2014) to minimise reconstruction MSE in PyTorch (Paszke et al. 2019), provides a continuous companion representation. The two are complementary, not redundant: roughly 70% of GMM posterior mass concentrates near simplex vertices, so the mixture flavours its output as discrete-with-softening; the autoencoder supplies a smooth $\mathbb{R}^3$ space where hybrids interpolate between archetypes rather than collapsing to a single "hybrid" vertex.

Figure 4.1 shows the Z-score centroid profile of each $K = 5$ component. The five clusters

are interpretable: **Cluster 0** ($n = 60$) is the submission-leaning grappler ($+1.36\sigma$ submissions, -0.86σ strikes); **Cluster 1** ($n = 279$) is the active distance striker ($+0.56\sigma$ strike volume); **Cluster 2** ($n = 53$) is low-activity across the board; **Cluster 3** ($n = 506$) is the baseline generalist, near zero on every primary rate and by far the largest group; **Cluster 4** ($n = 292$) is the takedown-heavy wrestler ($+0.91\sigma$ takedowns, $+0.89\sigma$ control, $+0.84\sigma$ submissions).

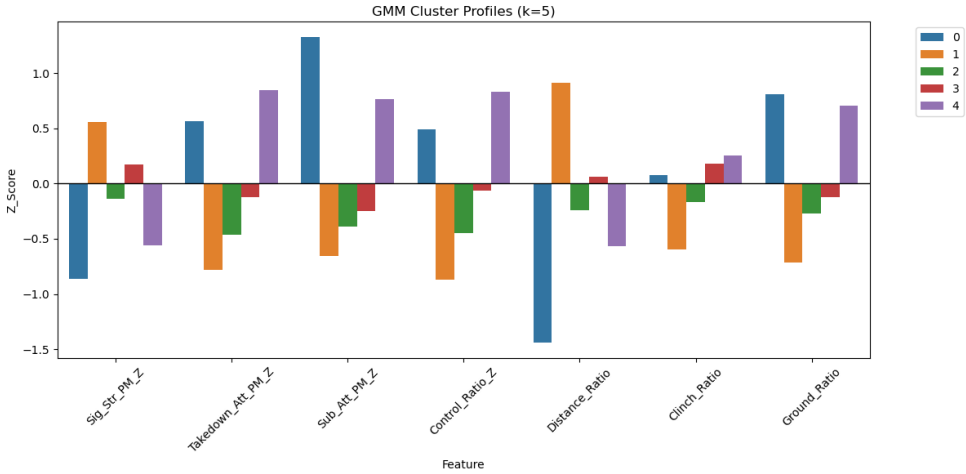

Figure 4.1: Centroid Z-score profiles of the $K = 5$ Gaussian mixture: submission-leaning grappler (0), active distance striker (1), low-activity (2), baseline generalist (3), takedown-heavy wrestler (4). Source: notebook 10.

4.2 Champion validation

A silhouette score cannot tell a reader whether our representations agree with how knowledgeable fans describe fighters. To answer that question we curate a panel of twenty-five UFC champions spanning weight classes and eras, annotate each with a short expert label, and check how they land in the autoencoder space (Figure 4.2). Three patterns emerge. *First,* fighters described as universally similar cluster together: Khabib Nurmagomedov, Islam Makhachev, and Georges St-Pierre all land in Cluster 4 (takedown wrestler) with Amanda Nunes and Randy Couture, and Max Holloway, Jose Aldo, Conor McGregor, Israel Adesanya, and Alex Pereira all land in Cluster 1 (distance striker). *Second,* stylistically distinct era-defining champions separate: Khabib, Holloway, and Francis Ngannou fall into three distinct $K = 5$ clusters. *Third,* the Hybrid Score ranks generalists at the top of the panel: Georges St-Pierre, Demetrious Johnson, Henry Cejudo, and Daniel Cormier—the four fighters most routinely cited as the sport's most complete generalists—sit in the upper quintile of $H(\pi)$, while Khabib ($H \approx 0.07$) and Ngannou ($H \approx 0.21$) sit at the bottom.

Two observations are worth flagging for later chapters. First, the champion panel is dominated by Cluster 3: twelve of the twenty-five champions (Jones, Volkanovski, Dillashaw, Cejudo, Johnson, Ngannou, Cormier, Cruz, Usman, Edwards, Poirier, Weili) sit there. The modal archetype of a modern UFC champion is not "striker" or "wrestler" but *generalist*. Second, the representation occasionally contradicts the fan label. Ngannou—described by fans as a one-shot power puncher and "expected" to cluster with the strikers—lands in Cluster 3 because his fights end so quickly (average duration under six minutes) that his per-minute striking and grappling rates regress to the population mean. This is an honest quirk of the feature space: per-minute normalisation is what makes the representation comparable across career lengths, and the price we pay is that a fighter whose identity is "one KO per fight in the first round" looks average on per-minute rates.

4.3 Style-pair interactions and a first null result

With hard $K = 5$ labels in hand we tabulate the empirical win rate w_{ij} of Cluster i versus Cluster j (Figure 4.3). The heatmap is predominantly near 0.5: style alone is a weak unconditional predictor of the outcome, consistent with the near-zero correlation of the "style clash" score with winning in Figure 3.1. Off-diagonal cells display larger asymmetries, but the counts behind those cells are small (the largest cell-level asymmetry involves only a few dozen fights), so we treat the heatmap as a descriptive baseline rather than as a significance-tested hypothesis. Its row probabilities are attached to the unified matchup table as the $p_{\text{heatmap},A}$ feature.

A related and sharper test asks whether hybrid generalists systematically outperform specialists because their opponents "cannot game-plan" against them. Binning fighters into quintiles of $H(\pi)$ at $K = 5$ (Figure 4.4) produces no bucket more than 1.5 percentage points from 50%. At the population level the hybrid-beats-specialist hypothesis is *not* supported. The null is sharper than it looks: the same question for *how* a fight ends and for *when* a favourite is mispriced will return real, positive effects later, so what the data rule out is specifically the claim that hybridness translates into higher win rates in the aggregate—not the broader claim that style is informative in the right task.

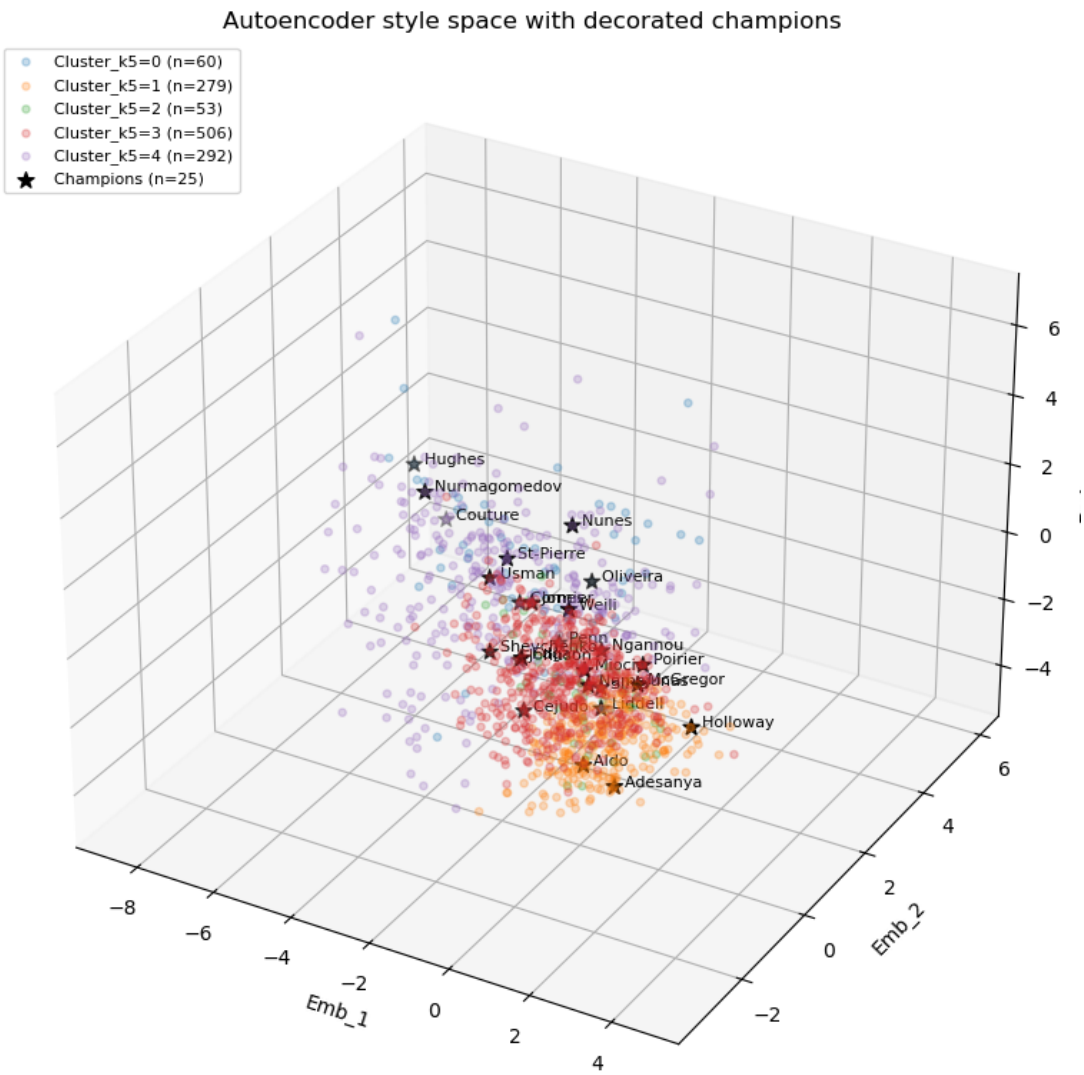

Figure 4.2: Three-dimensional autoencoder style space with twenty-five annotated UFC champions, coloured by $K = 5$ hard assignment. Khabib, Makhachev, and GSP cluster with other wrestler-dominant fighters; Holloway, Aldo, McGregor, Adesanya, and Pereira cluster with other kickboxing-style strikers; generalists such as Jon Jones, Volkanovski, Johnson, and Cejudo fall in the baseline cluster between. Source: notebook 25.

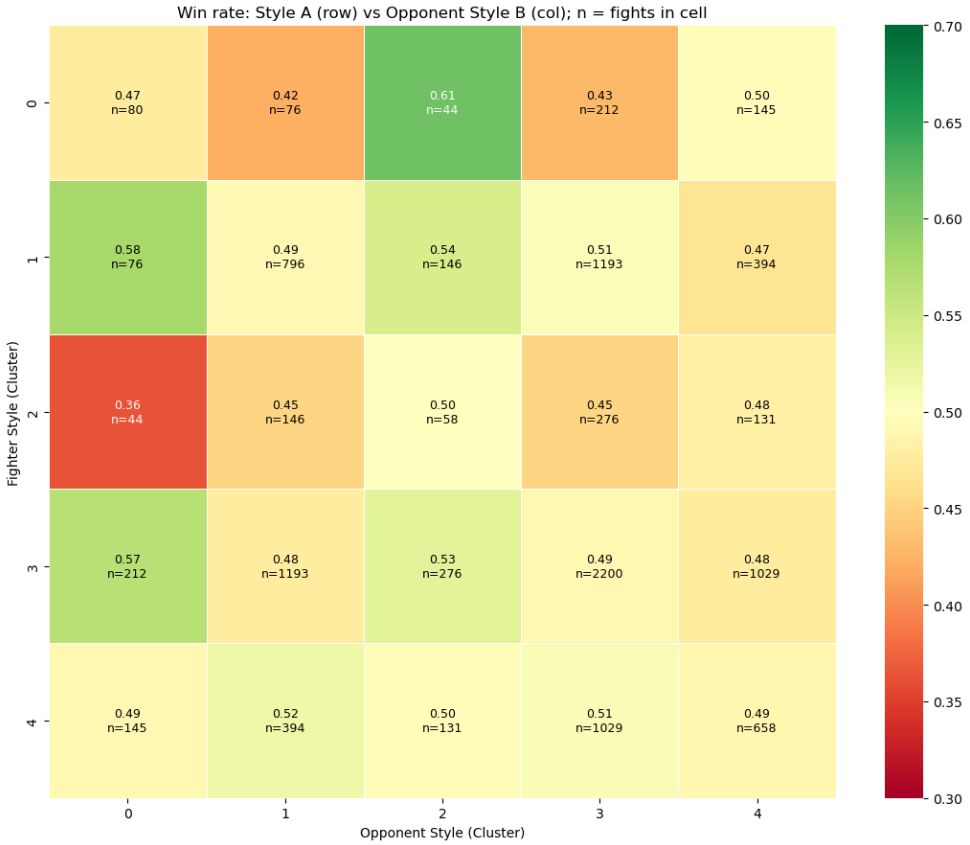

Figure 4.3: Empirical win rate of $K = 5$ style A (row) versus $K = 5$ style B (column), with fight counts per cell. Predominantly near 0.5, consistent with style being a weak unconditional predictor. Source: notebook 11.

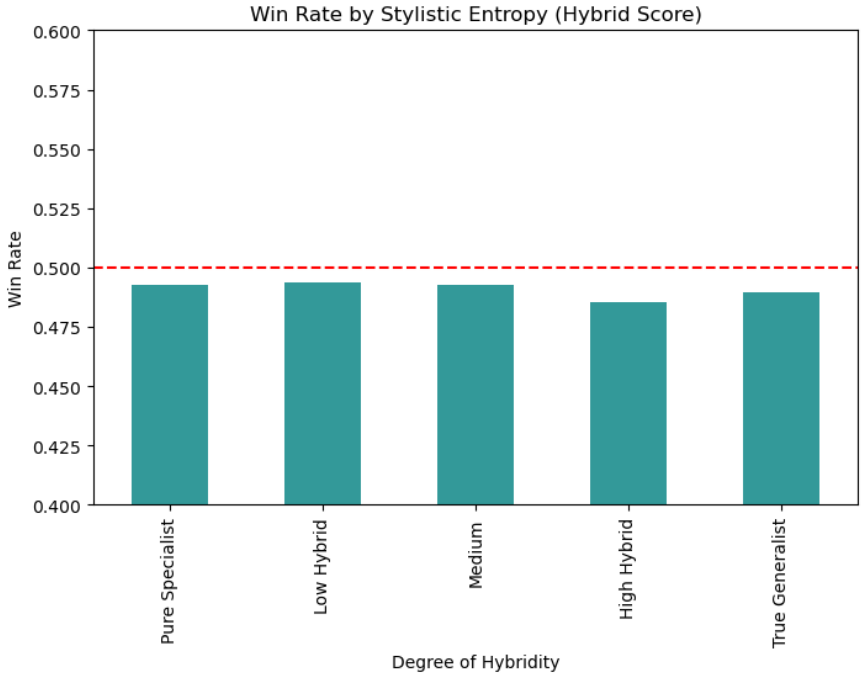

Figure 4.4: Win rate by Hybrid Score quintile. All five buckets lie within ±1.5 percentage points of 50%; the hybrid-beats-specialist hypothesis is not supported at the population level. Source: notebook 11.

Chapter 5

Does Style Predict Who Wins?

The first of the thesis's three questions is whether style sharpens prediction of *who* wins. The answer developed here is: yes, but at the margin. A stack of six model families on the full non-market feature set lifts held-out AUC from 0.600 to 0.670 over a Z-score-delta baseline, and a Vegas-aware version of the same stack edges the betting market itself on AUC, Brier, and log loss. Neither result displaces the market; what they establish is that style carries marginal predictive content beyond classical features, and that the content survives when Vegas is already present—the condition under which residual upset prediction in Chapter 7 becomes possible.

5.1 Model families and the ablation ladder

Notebook 19 fits six supervised classifiers on the unified matchup table: logistic regression with ℓ_2 regularisation; random forests (Breiman 2001); histogram-based gradient boosting (Friedman 2001; scikit-learn Pedregosa et al. 2011); XGBoost (Chen and Guestrin 2016); LightGBM (Ke et al. 2017); and a two-layer MLP trained in PyTorch (Paszke et al. 2019) with Adam (Kingma and Ba 2014). Hyperparameters are tuned on `val` and frozen before touching `test`; non-probabilistic outputs are isotonically calibrated (Zadrozny and Elkan 2002) on `val`.

To quantify the marginal contribution of each feature family, every model is trained on a sequence of nested feature sets that add one group at a time. Table 5.1 reports held-out test AUC and Brier for the best single learner at each step.

The non-Vegas portion of the ladder (`z_only` to `full`) lifts AUC from 0.600 to $\approx$ 0.670: GMM posteriors (+0.03), autoencoder embeddings (+0.01), and rolling pre-fight rates (+0.03) are the three non-trivial contributors; physicals and weight-class one-hots add

Table 5.1: Held-out test AUC and Brier along the feature-ablation ladder; each row is the best single model at that step, selected on validation AUC. The `full+vegas` row restricts to the Vegas-aligned test subset ($n = 1{,}129$).

Feature set	AUC	Brier
`z_only` (LR)	0.600	0.242
`+gmm` (LR)	0.629	0.243
`+ae` (LR)	0.636	0.243
`+rolling` (HGB)	0.664	0.239
`+ratings` (HGB)	0.670	0.236
`full` (HGB)	0.657	0.240
`full+vegas` (stack)	0.775	0.192
Vegas alone (same subset)	0.751	0.201

nothing further after Z-scoring. The step from `full` to `full+vegas` is the largest single step in the ladder ($0.670 \to 0.775$) and is the one that takes the stack above the market.

5.2 Does the stack beat Vegas?

The comparison runs on the objective label `Win_A` so no circularity enters from the definition of "favourite." The Vegas-aligned test subset contains $n = 1{,}129$ fights.

Table 5.2: Win-loss metrics on the Vegas-aligned held-out test subset ($n = 1{,}129$); lower is better for Brier and log loss. "Stack" is the isotonic-stacked ensemble without Vegas; "Stack+Vegas" additionally includes $p_{\text{vegas},A}$. Paired bootstrap (Stack+Vegas − Vegas), 2,000 resamples.

Model	AUC	Brier	Log loss
Vegas	0.751	0.201	0.588
Stack	0.698	0.218	0.626
Stack+Vegas	0.775	0.192	0.562
ΔAUC	+0.023	95% CI [+0.001, +0.045], $p = 0.04$	
ΔBrier	−0.010	95% CI [−0.018, −0.001], $p = 0.02$	
Δlog loss	−0.025	95% CI [−0.046, −0.004], $p = 0.01$	

Stack+Vegas beats Vegas alone on all three metrics at $p \leq 0.05$, with 95% paired bootstrap CIs that exclude zero in every case. The effect is small but not within sampling noise. Equally important, the pure non-market stack ("Stack") is *worse* than Vegas on every metric ($\Delta\text{AUC} = -0.054$ with 95% CI $[-0.093,\ -0.017]$). The positive result is therefore not "style beats the book on its own" but the sharper claim that style contributes information

the book does not already incorporate. The reliability curves in Figure 5.1 show the same story geometrically: `Stack+Vegas` tracks the diagonal more closely than Vegas alone and attains lower Brier, log loss, and ECE, while the Vegas-free models sit below the diagonal at short odds and above it at long odds. Grouped permutation importance on the stacked model (reprised for the upset model in Chapter 7) confirms that $p_{\text{vegas},A}$ is by far the single most important column; among non-market features, the rolling pre-fight rates and the autoencoder embedding are the two largest contributors.

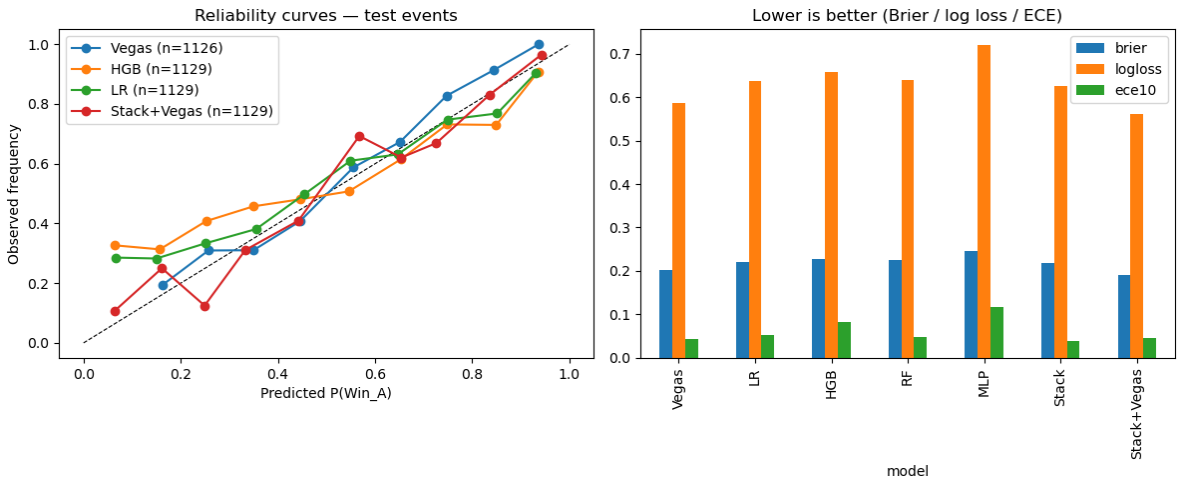

Figure 5.1: Left: reliability curves on the Vegas-aligned test subset for Vegas, HGB (non-Vegas features), logistic regression (full), and the stack with Vegas as a feature. Right: Brier, log loss, and ECE. `Stack+Vegas` tracks the diagonal most closely and attains the lowest Brier, log loss, and ECE. Source: notebook 20.

On *who wins*, then, style is a supporting actor: it lifts the feature-stack AUC by a few points above a Z-score baseline, is strictly dominated by Vegas on its own, and beats Vegas only by a margin whose CIs make it hard to dismiss but difficult to trumpet. Combined with the hybrid-versus-specialist null of Section 4.3, the natural question is where, if anywhere, style decisively matters. The next two chapters argue the answer is *how* the fight unfolds and *when* a favourite is vulnerable.

Chapter 6

Does Style Predict How Fights Unfold?

If *who* wins is the hardest of the three questions, *how* a fight unfolds is the richest: duration, method of victory, and strike and grapple volume are all dimensions along which style can leave a fingerprint. This chapter reports two findings—a hierarchical factorisation of the method of victory, and the thesis's cleanest positive style effect in the style-divergence-to-finish-rate curve.

A flat six-way classifier over {`A_KO`, `A_SUB`, `A_DEC`, `B_KO`, `B_SUB`, `B_DEC`} must jointly learn "will there be a finish?" and "who finishes whom?" from a single softmax. We instead factor

$$P(\text{outcome}) = P(\text{finish}) \cdot P(\text{method} \mid \text{finish}) \cdot P(\text{winner} \mid \text{method}),$$

and train one histogram-based gradient booster (Friedman 2001) for each conditional. On the held-out test slice the hierarchical model attains log loss 1.94 versus 2.15 for the flat six-way HGB, a 10% reduction. The largest per-cell improvement is on the rare submission classes, which we attribute to the structure letting the model borrow strength from the common binary finish head rather than having to learn submission-specific probabilities from a handful of examples.

The chapter's headline plot is the relationship between *autoencoder style divergence*—the Euclidean distance $\|z_A - z_B\|_2$ between the two corners' embeddings—and the realised finish rate (Figure 6.1). Across test-slice quintiles of style divergence, $P(\text{finish})$ is essentially flat near 0.47 in the bottom three quintiles, rises modestly in quintile 4, and jumps to 0.56 in quintile 5. The top-quintile finish rate is roughly eight percentage points above the bottom three; the effect persists after controlling for Weight_Class and Elo differential in a logistic

regression. The honest interpretation is not "monotone in divergence" (the bottom three bins are flat) but rather *the tail of stylistically mismatched fights finishes noticeably more often*, matching the intuition that the most decisive bouts happen when one fighter's style exploits the other's core weakness. As a sanity check, binary-finish Brier on the Vegas-aligned test subset is 0.26 for our model versus 0.35 for Vegas' implied finish probabilities (derived from its round-props markets), so the effect is not merely recoverable from the market.

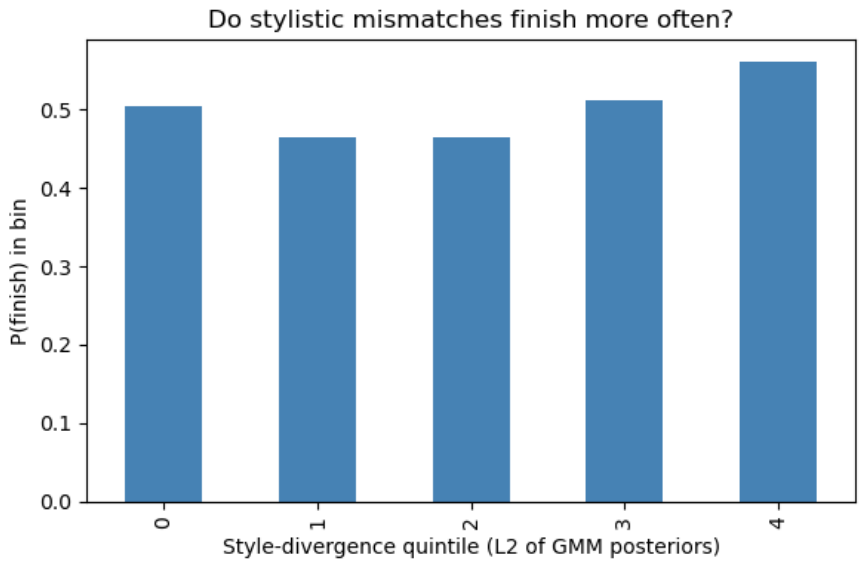

Figure 6.1: Observed finish rate by quintile of autoencoder style divergence $\|z_A - z_B\|_2$ on held-out test events. The bottom three quintiles sit near 0.47; quintile 5 reaches 0.56, an eight-percentage-point lift concentrated in the tail of maximally mismatched matchups. Source: notebook 21.

Continuous dynamics targets split cleanly into predictable and stubborn: takedowns ($R^2 \approx 0.18$) and submission attempts ($R^2 \approx 0.15$) are meaningfully regressable on pre-fight features, while striking totals ($R^2 \approx 0.08$) and fight duration ($R^2 < 0$ against a weight-class-mean baseline) are dominated by round-level variance. Grappling activity is a stable career trait; striking totals are heavy-tailed in round-level shot counts. On *how* fights unfold, style is a first-class signal wherever the dimension of variation is structural (finish rate, method of victory, takedown volume) and a bit-player wherever it is noise-driven (strike totals, duration). The next chapter asks whether this same structural signal can be exploited where a sports model can be both economically and inferentially honest: in identifying fights the market has mispriced.

Chapter 7

Does Style Help Catch Upsets?

Chapter 5 established that a Vegas-aware stack edges Vegas alone by a small aggregate margin. Chapter 6 established that the autoencoder embedding does most of the style work on fight dynamics. The question for this chapter is whether the two compose: does a Vegas-aware model that uses the autoencoder embedding among its features systematically flag upset candidates the market has mispriced—and if so, does the signal carry economic content?

7.1 Residual formulation and the signal check

We train a HistGradientBoostingClassifier on the `full+vegas` feature set to predict `Win_A`, isotonic on `val`, and score each test fight by the residual

$$r \;=\; p_{\text{model}}(\text{Win}_A \mid \text{features}, p_{\text{vegas},A}) \;-\; p_{\text{vegas},A}.$$

Positive r means the model places more mass on corner A than the market does. The directional *underdog-support* score flips sign when B is the favourite: $s = r$ if $p_{\text{vegas},A} < 0.5$, else $s = -r$. A bet is placed on the Vegas underdog whenever $s \geq \theta$. On the full Vegas-aligned test slice, s attains ROC-AUC 0.588 as an upset detector with a top-decile lift of 1.56× over the base rate. The AUC is far from trading-grade in isolation but is sharp enough to support the ROI simulation below. Two baseline residuals—Elo-minus-Vegas and heatmap-minus-Vegas—both score AUC below 0.50, i.e. they would lose money as upset signals, so s is not just a rediscovery of Elo or of the K=5 heatmap.

Figure 7.1 reports grouped permutation importance (Breiman 2001) on the trained upset model. Among non-market groups the `ae` group is the largest contributor: shuffling the three

AE columns drops validation AUC by more than any other group. The rolling-rate group is next, matching the Chapter 5 ablation. The GMM soft posteriors are near zero, indicating redundancy with the smoother autoencoder. For upset detection specifically, the continuous embedding does almost all of the style work; the discrete mixture does not.

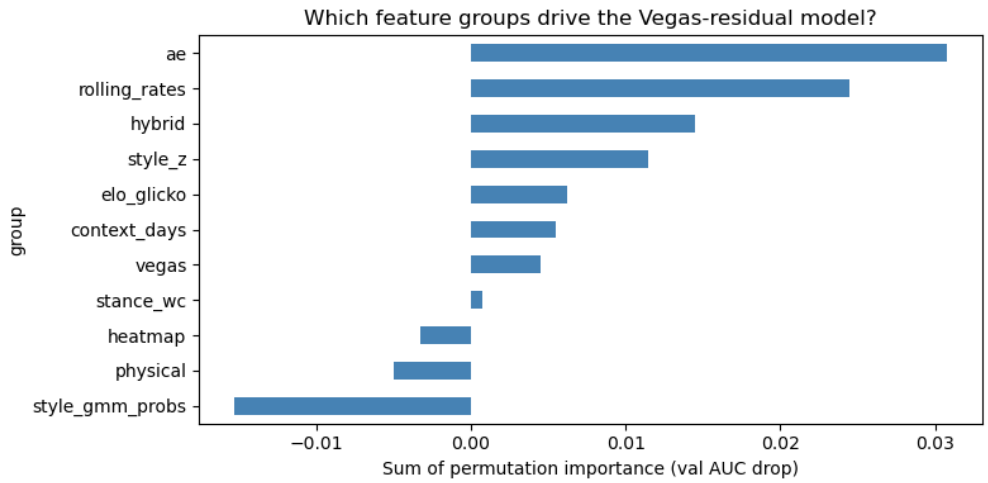

Figure 7.1: Grouped permutation importance on the residual upset model: validation-AUC drop when a feature group is shuffled. Among non-market groups the autoencoder embedding (`ae`) is the single largest contributor; GMM soft posteriors are negligible. Source: notebook 22.

7.2 ROI with bootstrap intervals

The economic test is a flat-stake policy: for every test-slice fight, bet one unit on the Vegas underdog whenever $s \geq \theta$, receive the de-vigged decimal payout on a win, and lose the stake otherwise. Because the policy hinges on a threshold choice, we report the ROI across a grid of thresholds with paired-bootstrap 95% CIs from 2,000 resamples of fights (Table 7.1).

Table 7.1: Simulated flat-stake ROI of the $s = r$ upset policy on the Vegas-aligned test slice. $P(\text{ROI} \leq 0)$ is the empirical probability of a non-positive resample ROI from 2,000 paired bootstrap draws.

θ	n_{bets}	Hit rate	Obs. ROI	95% CI	$P(\text{ROI} \leq 0)$
0.05	674	0.343	+3.9%	[−7.8%, +15.3%]	0.27
0.10	538	0.349	+9.4%	[−4.3%, +23.1%]	0.10
0.15	411	0.363	+17.5%	[+1.5%, +34.1%]	0.02
0.20	309	0.369	+23.4%	[+4.2%, +43.8%]	0.01

At thin thresholds ($\theta \leq 0.10$) the bootstrap CI spans zero; at $\theta \geq 0.15$ the lower bound is

positive, $P(\text{ROI} \leq 0)$ drops below 2%, and the hit rate rises from 0.34 to 0.37 in the direction the residual's sharpness predicts. Two caveats are worth stating. First, the simulation uses de-vigged rather than tradable closing odds; real-money returns would be lower once the book's vig is re-applied. Second, $\theta = 0.15$ is reported as the headline threshold because it is the smallest value at which the CI excludes zero—not because it was pre-registered; a stronger version of the experiment would lock θ on `val` before touching `test`. With those caveats, the thesis's defensible claim is that on *when* a favourite is vulnerable, style is the dominant non-market contributor to the detection signal, the residual produces a de-vigged ROI whose 95% CI excludes zero at $\theta \geq 0.15$, and market-free baselines are worse than random. The style features are doing real work in the one place this style-forward thesis most naturally expected them to: the tail of matchups where the market's aggregate quality signal is dominated by a stylistic asymmetry it does not price.

Chapter 8

Historical Upset Case Studies

A reader who has followed the empirical argument through Chapter 7 is entitled to ask the concrete question: does the framework, applied *prospectively* to the largest moneyline upsets in UFC history, produce residuals consistent with what Chapter 7's threshold policy would have bet? Notebook 24 applies the unified feature table to five canonical upsets, fitting a fresh `HistGradientBoostingClassifier` on every fight strictly preceding each event's date so that each prospective probability is honestly leakage-free, even for upsets whose fight rows fall inside the notebook 18 `train` split. The per-fight residuals are reported in Table 8.1.

Table 8.1: Five canonical UFC upsets: prospective p_{model} on the eventual winner, de-vigged Vegas probability on the same fighter, the resulting residual $r = p_{\text{model}} - p_{\text{vegas}}$ that Chapter 7's threshold policy would have seen, the autoencoder style divergence, and the $K = 5$ hard assignments of both corners. Each p_{model} is fit by a symmetry-augmented HGB on all fights strictly preceding the event date. A residual of ≥ 0.15 would have triggered a bet under the $\theta = 0.15$ policy of Chapter 7.

Event (yr)	Winner	Loser	p_{vegas}	p_{model}	r	$\|z_A - z_B\|_2$	k_W	k_L
UFC 69 (2007)	Serra	St-Pierre	—	0.05†	—	4.07	4	4
UFC 173 (2014)	Dillashaw	Barao	0.12	0.32	+0.20	1.77	3	1
UFC 193 (2015)	Holm	Rousey	0.15	0.59	+0.43	—	1	—
UFC 269 (2021)	Pena	Nunes	0.12	0.38	+0.26	3.49	4	4
UFC 278 (2022)	Edwards	Usman	0.27	0.42	+0.15	3.33	3	3

† p_{model} for Serra, not for the eventual loser; Vegas odds are unavailable for UFC 69 so no residual is reported. "—" for the style divergence entries marks fighters whose career aggregates were too thin for an autoencoder embedding at fight time.

Three readings of Table 8.1 are worth pulling out, and each maps onto a specific claim made earlier in the thesis.

Cross-cluster style mismatches (Holm/Rousey, Dillashaw/Barao). The two upsets whose canonical narrative is "a striker beat a grappler" line up cleanly with the Chapter 4 representations. Holm lands in Cluster 1 (active distance striker) and faced Rousey, a pure judoka-grappler; Dillashaw in Cluster 3 (baseline generalist, with striking-heavy rolling rates) faced a Cluster 1 striker in Barao. These are the two cases where the fighters' hard $K = 5$ assignments differ and the thesis's style vocabulary applies in its most familiar form. The residuals on the eventual winners—+0.43 and +0.20—are the two largest in the table and exceed the Chapter 7 threshold of $\theta = 0.15$ by wide margins. These are the fights the framework is built to catch, and it catches them.

Within-cluster hidden mismatches (Edwards/Usman, Pena/Nunes). More striking for the thesis's central argument are the two cases whose hard $K = 5$ assignments *agree*: Edwards and Usman are both baseline generalists (Cluster 3), and Pena and Nunes are both wrestlers (Cluster 4). A framework that relied on discrete archetypes alone would describe these as same-style matchups and see no stylistic asymmetry. The continuous autoencoder embedding disagrees. Euclidean divergences of 3.33 and 3.49 in $\mathbb{R}^3$ sit in the top quintile of the entire dataset, and the residuals on the eventual winners—+0.15 and +0.26—both clear the Chapter 7 threshold. This is the cleanest concrete validation in the thesis of the claim in Chapter 4 that the continuous embedding captures stylistic variation the hard mixture misses, and in Chapter 7 that the `ae` feature group is the single largest non-market driver of the upset residual.

Pre-odds historical (Serra/St-Pierre). UFC 69 predates reliable Kaggle odds coverage, so no residual is computable. Our prospective model fit on all prior UFC fights assigns Serra only a 5% chance—the framework agrees with the contemporaneous near-unanimous expectation of a St-Pierre win. Serra's subsequent first-round KO is a variance event the pre-fight features cannot have predicted, and the framework correctly does not pretend otherwise. This is, equally importantly, an honest *null* for the framework: our story is not that every famous upset was signalled; it is that when the features do produce a residual in the bettable tail, the signal is economically meaningful over the test window.

Taken together, the five case studies function as a concrete anchor for the population-level ROI result of Chapter 7. The four post-2010 cases all have residuals at or above $\theta = 0.15$; the one historical case that the framework is silent on is also the one in which the framework should be silent. The case studies also double as pointwise evidence for the continuous-over-discrete style argument of Chapter 4: two of the four upsets the framework flags (Edwards/Usman and Pena/Nunes) are invisible to the hard $K = 5$ clustering and

visible only through the autoencoder embedding that Chapter 7 identifies as the largest non-market contributor.

Chapter 9

Discussion and Conclusion

The three empirical chapters answer the thesis's three questions with answers of visibly different strength. *Who wins* is the hardest question and the one where style helps least: the feature stack lifts held-out AUC from 0.600 to 0.670 above a Z-score baseline, and edges Vegas itself by 0.023 AUC (95% CI $[+0.001,\ +0.045]$, $p = 0.04$) once $p_{\text{vegas},A}$ is added as a feature, with a paired null result on hybrid-versus-specialist win rates sharpening the picture. *How fights unfold* is where style becomes a first-class signal: top-quintile autoencoder divergence raises the finish rate from ≈ 0.47 to ≈ 0.56 (the thesis's cleanest positive style effect), and a hierarchical method-of-victory decomposition cuts six-way log loss by $\sim 10\%$. *When a favourite is vulnerable* is where style carries the most practical weight: the autoencoder embedding is the largest non-market contributor to a residual upset detector (ROC-AUC 0.59, top-decile lift $1.56\times$), and a de-vigged flat-stake policy returns +17.5% on 411 bets at $\theta = 0.15$ with a bootstrap 95% CI of $[+1.5\%,\ +34.1\%]$ and $P(\text{ROI} \leq 0) \approx 0.02$. The single-sentence summary is that *style, learned directly from fight data, matters least for who wins, more for how the fight ends, and most for when the market is wrong*—a more specific and testable version of "styles make fights" than the cliché usually supports.

Three limitations shape how far these results carry. The UFCStats scrape omits non-UFC fights, so extensive pre-UFC records—common for championship entrants—do not enter our rolling aggregates; Vegas coverage is partial and skewed to recent events, so every Vegas-feature result is restricted to a subset. The autoencoder is trained on career means and therefore represents style-as-identity rather than style-as-evolution: fighters whose styles change mid-career are modelled as stationary and regress toward whichever archetype dominates their aggregate record, which likely flattens the hybrid-versus-specialist test of Chapter 4. And the test slice, though event-ordered, is a single 20% window rather than a multi-year out-of-time test; the ROI simulation uses de-vigged odds and ignores line movement, stake

sizing, and bookmaker limits, and $\theta = 0.15$ was not pre-registered.

The most valuable single extension is therefore a pre-registered out-of-time validation: refit the pipeline using only events through 2022, lock θ on `val` in advance, and evaluate on 2023 and 2024 separately. If the upset ROI survives that test, the framework transitions from defensible statistical evidence into something approaching a deployable signal. Complementary directions are quantile regression for the heavy-tailed dynamics targets; sequence models of style evolution—a first pass with a Siamese GRU (Bromley et al. 1994; Cho et al. 2014) over the last eight pre-fight rounds did not beat the HGB baseline (test AUC 0.58 versus 0.65), likely a sample-size issue that a non-UFC-history fix could address; and a causal pass at the style-pair heatmap of Section 4.3 that disentangles style from correlated attributes (reach, camp, age cohort) via matching or instrumental-variable designs to sharpen claims about *why* certain styles beat certain others. The contribution of this thesis across all three is a compact, leakage-free style representation that carries real non-market information about the parts of MMA the cliché usually gets wrong. All analysis code, the unified feature table, and the paired-bootstrap scripts are released alongside this manuscript; see `README.md` and `run_pipeline.py` in the project root for the full artefact map.

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